```
D:\Software\Python\Python311\python.exe E:\PythonProjectsByYear\Year2025\MC-PIKAN_Official\experiments\Volterra1D_MCPI.py
The network model is PIKAN
Layers: ModuleList(
  (0): ChebyKANLayer(in_features=1, out_features=10, degree=5)
  (1): ChebyKANLayer(in_features=10, out_features=10, degree=5)
  (2): ChebyKANLayer(in_features=10, out_features=1, degree=5)
)
Total trainable parameters: 720
Model on device: cuda:0
Start training...
Max Epochs: 2000, Learning Rate: 0.001, Epsilon: 1e-20, Optimizer: Adam
Training: 100%|██████████| 2000/2000 [00:09<00:00, 220.33epoch/s, Loss=1.29e-06, LR=1.0e-03, Best=1.29e-06]
Training finished. Best loss: 1.11e-06. Model saved to results/best_model.pth
Loss history saved to results/loss_history.npy
Training time: 9.89218282699585s
Relative loss: 0.0018984471680599647
L2 loss: 0.06001915782690048
Parameter containing:
tensor([[[ 0.0015, -0.0211,  0.3154,  0.0671,  0.1327, -0.2262],
         [-0.1233, -0.0247,  0.0264,  0.2096,  0.1220, -0.0433],
         [-0.0352,  0.0738,  0.0403,  0.3578,  0.0906, -0.1600],
         [-0.2908, -0.0860, -0.4678,  0.0278,  0.1993,  0.0595],
         [-0.0180,  0.1188,  0.1252, -0.1438,  0.1200,  0.2226],
         [-0.0218,  0.2284,  0.2940,  0.4029,  0.2123, -0.1680],
         [-0.0025,  0.0397,  0.2492, -0.0668, -0.0491,  0.1224],
         [-0.4666,  0.0635,  0.0508,  0.1692, -0.1068,  0.1438],
         [ 0.0345, -0.0674,  0.0264, -0.1679, -0.0501, -0.0568],
         [-0.0981,  0.0701, -0.0297,  0.0225,  0.1587, -0.0186]]],
       device='cuda:0', requires_grad=True)
Parameter containing:
tensor([[[ 3.5835e-02,  4.3597e-02, -3.2066e-02, -2.8146e-02,  1.4577e-02,
           3.9802e-02],
         [-1.4049e-03, -1.3495e-02,  1.3313e-02,  8.8128e-03, -2.5578e-03,
          -4.1426e-02],
         [ 4.1543e-02,  2.0691e-02, -9.4714e-03, -7.3274e-02, -1.7464e-03,
           5.3575e-02],
         [-1.0738e-02,  2.5562e-02, -1.6177e-02, -1.7965e-02, -9.3052e-03,
           4.4808e-02],
         [-1.1526e-02,  3.0299e-02,  2.7470e-02, -1.5887e-02,  1.9832e-02,
           4.3969e-02],
         [-7.1993e-03, -3.4418e-02,  3.5335e-02, -1.0537e-03, -1.3415e-02,
          -2.8822e-02],
         [-1.9653e-02, -4.3714e-03,  7.9978e-03, -4.5955e-02, -4.0380e-02,
          -5.6790e-03],
         [-3.0913e-02, -3.8797e-02, -3.6207e-03,  5.1079e-02,  4.1235e-02,
          -9.2516e-02],
         [-9.5518e-03,  1.5013e-02, -2.9706e-02, -1.1783e-02,  2.7821e-02,
           3.5061e-02],
         [-4.3198e-03, -8.4740e-02,  9.7127e-03,  4.2258e-02, -2.3149e-03,
          -4.6796e-02]],
```

```
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          -3.8387e-02],
         [ 5.0602e-03,  2.1167e-02,  6.2762e-03,  2.6286e-02,  3.2144e-02,
           1.7645e-02],
         [-1.9601e-02,  4.1045e-02,  1.0004e-02, -2.4471e-02, -1.0165e-02,
          -7.6924e-03],
         [-3.8617e-02, -2.1620e-02,  1.8359e-02, -1.2443e-03, -5.9233e-02,
           1.7056e-02],
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           4.2973e-02],
         [-2.5497e-02, -3.4922e-02,  4.9436e-02, -7.0054e-03, -3.6660e-02,
          -3.4699e-02],
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           1.5507e-02],
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          -3.3406e-02],
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           2.7650e-02],
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```

```
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```

```
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         [-5.5206e-02, -4.7926e-03,  2.3705e-02, -1.4668e-02, -4.6467e-02,
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Parameter containing:
tensor([[[ 0.0349, -0.0202, -0.0330, -0.0427,  0.0247,  0.0369]],

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        [[ 0.0287,  0.0189, -0.0377, -0.0114,  0.0255, -0.0021]],

        [[-0.0117, -0.0226, -0.0220,  0.0099,  0.0044, -0.0642]],

        [[ 0.0256, -0.0142, -0.0095,  0.0458,  0.0220, -0.0489]],

        [[ 0.0267, -0.0121, -0.0203,  0.0135,  0.0515, -0.0395]],

        [[ 0.0190, -0.0277,  0.0102,  0.0103, -0.0292, -0.0098]],

        [[ 0.0080,  0.0736, -0.0158, -0.0202,  0.0224,  0.0564]],

        [[ 0.0077, -0.0218, -0.0284, -0.0387,  0.0354,  0.0206]],

        [[ 0.0356,  0.0279, -0.0292,  0.0196,  0.0100, -0.0347]]],
       device='cuda:0', requires_grad=True)
=========================================================================================
======
Layer (type:depth-idx)                   Output Shape              Param #
```

```
==========================================================================================
CPIKANModel                                [1, 1]                    --
├─ModuleList: 1-1                           --                        --
│    └─0.weight                                                        ├─60
│    └─1.weight                                                        ├─600
│    └─2.weight                                                        └─60
│    └─ChebyKANLayer: 2-1                    [1, 10]                   60
│    │    └─weight                                                      └─60
│    └─ChebyKANLayer: 2-2                    [1, 10]                   600
│    │    └─weight                                                      └─600
│    └─ChebyKANLayer: 2-3                    [1, 1]                    60
│    │    └─weight                                                      └─60
==========================================================================================
Total params: 720
Trainable params: 720
Non-trainable params: 0
Total mult-adds (Units.MEGABYTES): 0.00
==========================================================================================
Input size (MB): 0.00
Forward/backward pass size (MB): 0.00
Params size (MB): 0.00
Estimated Total Size (MB): 0.00
==========================================================================================

进程已结束，退出代码为 0

```